Chapter 7

Case histories

This chapter presents some applications of surface wave tests for site characterization. The aim is to give an overview of the processes described in previous chapters with some examples, which will be presented step by step. First, simple datasets will be presented, for which it is possible to compare different processing and inversion techniques. This is not always the case; in some circumstances, it is not easy to process the experimental data because of several reasons:

- Inadequate signal-to-noise ratio
- Influence of higher modes, which are difficult to identify
- Influence of body waves
- Discontinuous retrieval of the dispersion curve over the frequency range of interest
- Ambiguities due to lack of resolution and spatial aliasing

These aspects necessitate some experience and engineering judgment and can cause the results to be less reliable (increasing relevance of the nonuniqueness of the solution). It is almost impossible to provide a comprehensive picture of complex datasets, and there is no substitute for experience and engineering judgment in the analysis of experimental data. Accurate design of the experimental setup and the use of adequate equipment and field procedures are prerequisites for obtaining reliable results.

The first section of the chapter provides an example of surface wave analysis of active-source data for the evaluation of the shear wave velocity profile, comparing different processing approaches. The inverse problem is subsequently solved, and there is a discussion of the relevance of a *priori* assumptions on model parameters, with specific references to the existence and position of a water table.

Section 7.2 reports an example of a combined dataset for which active-source data and passive-source data are available. The combination allows an extension of the frequency range from which the experimental dispersion curve can be retrieved and hence the evaluation of the shear

wave velocity profile to larger depths, with no need for heavy and expensive active sources. Several approaches for the solution of the inverse problem are compared for this dataset.

In Section 7.3, two examples of surface wave analysis for the evaluation of the shear wave velocity profile and the shear damping ratio profile are reported.

In Section 7.4, a case in which higher modes of Rayleigh wave propagation play a relevant role is presented to discuss the interpretation of more complex datasets.

Surface wave data also can be extracted from datasets that have been collected for other purposes. In Section 7.5, the exploitation of the so-called *ground-roll* in seismic reflection records for the analysis of surface is presented using two case histories.

7.1 COMPARISON AMONG PROCESSING TECHNIQUES WITH ACTIVE-SOURCE METHODS

The test site is located in Saluggia in the northern part of Italy, close to the Dora Baltea River; it is part of a large flat area of fluvial sediments. The subsoil is basically composed of gravels and gravelly sands, with the presence of fine sand and clayey silt in the form of lenses. The water table is very shallow, between 2 and 3 m below the ground surface. The results of a cross hole (CH) test at the site are available from a previous geotechnical survey. Details on the site are reported by Foti (2000).

Data have been collected using 24 vertical geophones (4.5 Hz natural frequency) with two different setups having receiver spacing equal to 1 and 3 m, respectively, and with the source-first geophone offset equal to the inter-receiver spacing. Impact sources have been used for both testing configurations: a 6 kg sledgehammer and a 130 kg weight-drop system height 3 m above the ground level respectively. The two configurations have been chosen to investigate the high- and the low-frequency ranges, respectively.

Examples of the experimental data are reported in Figure 7.1, which shows the vertical stacking of the 15 shot gathers recorded for each of the test setups. The difference in frequency content is evident in the seismograms: the central frequency of the trace at 3 m from the shot is about 25 Hz with the weight drop and 50 Hz with the sledgehammer. Also, a higher mode appears with the larger relative amplitude in the short offset gather with the lighter source.

7.1.1 Two-station (spectral analysis of surface waves)

For the two-station procedure, the signal pairs at the following inter-receiver distance have been used: 3, 6, 12, 18, and 30 m. Figure 7.2 shows an example of the spectral quantities needed for the phase velocity evaluation (see Section 4.3). A set of traces from seven repetitions of the impact has been

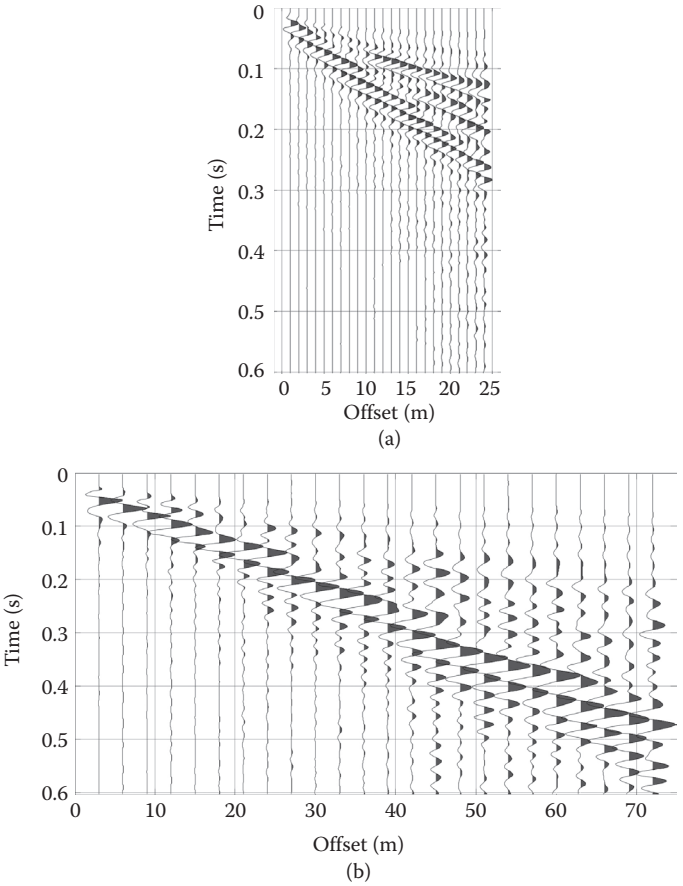


Figure 7.1 Experimental data: (a) 1 m spacing and 6 kg sledgehammer; (b) 3 m spacing and 130 kg weight-drop system (from 3 m height).

used to calculate all the average quantities. The cross-power spectrum phase is used to estimate the frequency-dependent time delay among the receivers and, because the inter-receiver distance is known, the phase velocity. The other quantities are used to locate the distribution of energy (through the autopower spectrum at the two receivers) and the frequency ranges with a high signal-to-noise ratio (SNR) (corresponding to a coherence function close to 1). This information is used to recognize the frequency range over which the cross-power spectrum is reliable. The useful frequency range selected for this pair of receivers goes from about 10 to 35 Hz.

A crucial aspect is related to phase unwrapping; indeed, the low SNR makes it difficult to clearly identify possible phase jumps in the frequency range below 10 Hz. An error in unwrapping the phase would badly affect

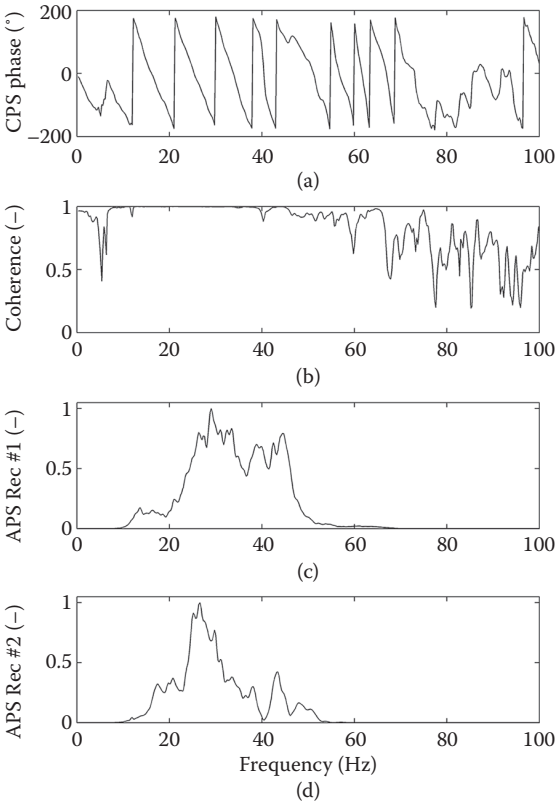


Figure 7.2 Example of signal processing for the two-station SASW procedure (source: weight-drop system; receiver spacing and source to receiver distance: 18 m): (a) phase of the cross-power spectrum; (b) coherence function; (c) autopower spectrum for receiver #1; (d) autopower spectrum for receiver #2.

the evaluation of the dispersion curve. A careful inspection of the comparison of the branch of the experimental dispersion curve obtained with different spacing is of paramount importance. Figure 7.3 shows the different branches of the experimental dispersion curve as retrieved from the different receiver couples. A possible strategy for data processing starts from closely spaced receiver pairs, which typically provide reliable information in the high-frequency range with high SNR. Then, when analyzing data from widely spaced receiver couples, it is possible to check the consistency among the different branches of the experimental dispersion curve to identify possible errors due to phase unwrapping. With good quality data, a significant overlap of information from different receiver pairs is typically obtained. In any case, the evaluation of the experimental dispersion curve requires experience and engineering judgment.

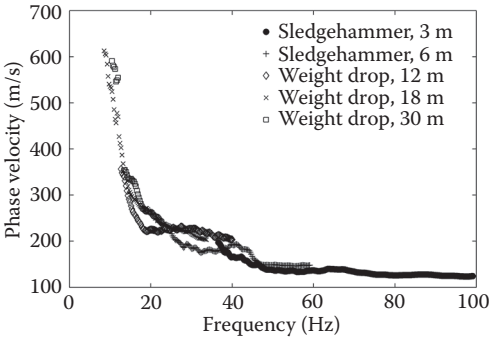


Figure 7.3 Two-station SASW experimental dispersion curve obtained from the analysis of several receiver pairs extracted from the dataset of Figure 7.1.

7.1.2 Frequency–wavenumber analysis

The analysis of multistation data is typically performed using transform-based methods in which the experimental data are transformed from the time–space domain in which they have been collected into a domain in which phase shifts associated with the propagation can be identified. In Figure 7.4a, the amplitude of the frequency–wavenumber spectrum of the experimental data of Figure 7.1b is reported. As detailed in Section 4.6.1, several options are available for the evaluation of the transform. In this case, the spectrum has been obtained with the application of a double Fourier transform. Although this is likely the simplest option, its application is limited to datasets collected with equally spaced receivers. The typical acquisition parameters allow for a sufficient resolution in the frequency domain because this is associated with the length of the acquisition window, which is constrained by the necessity of also recording the whole wave train at the farthest receiver. However, the resolution in the wavenumber domain is limited because the number of available receivers and logistic space constraints typically limit the length of the array. For this reason, a zeropadding procedure is necessary in space domain to obtain an adequate resolution in the wavenumber domain. For example, the spectrum in Figure 7.4a has been evaluated with a zero padding to 1024 sample points in space (i.e., 1000 signals with zero amplitude have been appended to the experimental dataset). The experimental dispersion curve is evaluated from the location of spectral density maxima in Figure 7.4a. From this dataset, the dispersion curve is retrieved consistently in the frequency range between 8 and 43 Hz. The limitation in the high-frequency range is imposed by the length of the array and by receiver spacing. Indeed, on the one hand, the attenuation of high-frequency components prevents the possibility of getting high SNR for the farthest

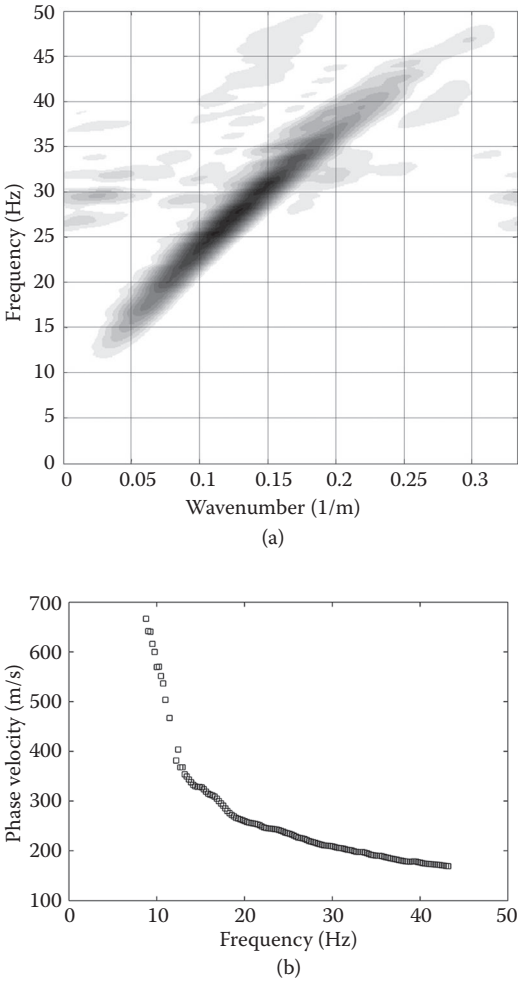


Figure 7.4 Analysis in the *f–k* domain of experimental data from Figure 7.1b: (a) amplitude of the frequency–wavenumber spectrum; (b) experimental dispersion curve evaluated from the location of the maxima of the spectrum.

receivers; on the other hand, receiver spacing limits the maximum available wavenumber.

On a complementary basis, the experimental data from the 1 m spacing array (Figure 7.1a) allow the experimental dispersion curve to be retrieved in the high-frequency range (Figure 7.5). The agreement between the two experimental datasets in the frequency range in which the relative

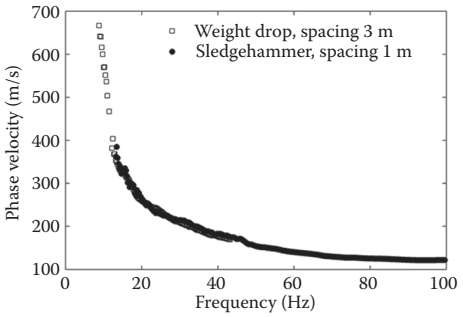


Figure 7.5 Experimental dispersion curve obtained from the combination of the two datasets of Figure 7.1 with *f–k* analysis.

dispersion curve overlaps is very good, showing that the same experimental mode is retrieved.

A key advantage of multistation processing is that, in principle, the dispersion curve is obtained from the location of the maxima without a need for engineering judgment on the part of the operator. For more complex datasets, the influence of higher modes and presence of frequency ranges with unclear maxima due to low SNR lead to the necessity of some engineering judgment for the evaluation of reliable branches of the dispersion curve.

As seen in Section 4.6, other transform-based methods are available. A tool widely used is the slant-stack or τ-p transform, which allows for the set of signals to be represented as the superposition of straight-line events. The two steps of the *f–p* transform of the experimental data in Figure 7.1b are shown in Figure 7.6. On the basis of the subsequent application of the τ-p transform (Figure 7.6b) and a 1D Fourier transform (Figure 7.6c), it is possible to transform the data in the frequency–slowness domain where the experimental dispersion curve is retrieved as the location of the maxima.

A similar procedure is implemented in the multichannel analysis of surface waves (MASW) method proposed by Park et al. (1999). For the same dataset of Figure 7.1b, the frequency–wavenumber, frequency–slowness, and frequency–velocity spectra are plotted in Figure 7.7. The frequency–velocity domain is obtained straightforwardly from the frequency–slowness domain because the velocity is the inverse of the slowness. Its use has the advantage that the experimental dispersion curve is directly visualized as the spectral maxima.

A comparison of the average experimental dispersion curve obtained with the two-station spectral analysis of surface waves (SASW) procedure

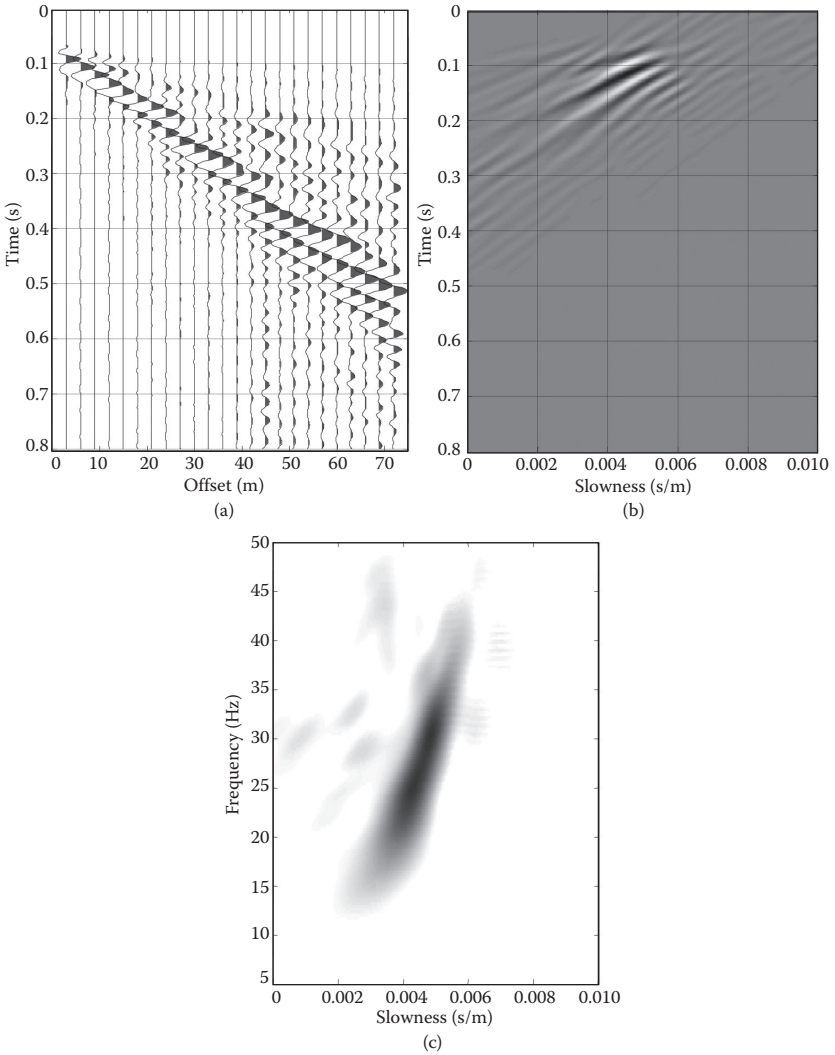


Figure 7.6 Analysis in the frequency–slowness domain of experimental data of Figure 7.1b. (a) experimental data, (b) τ-p spectrum, and (c) *f–p* spectrum.

with the experimental dispersion curves obtained with multistation procedures is reported in Figure 7.8, showing that the different procedures lead to comparable results in the same dataset. In particular, the dispersion curves obtained with frequency–wavenumber analysis and frequency–slowness analysis are indistinguishable.

The inversion of the experimental dispersion curve obtained from the analysis in the frequency–wavenumber domain is reported below.

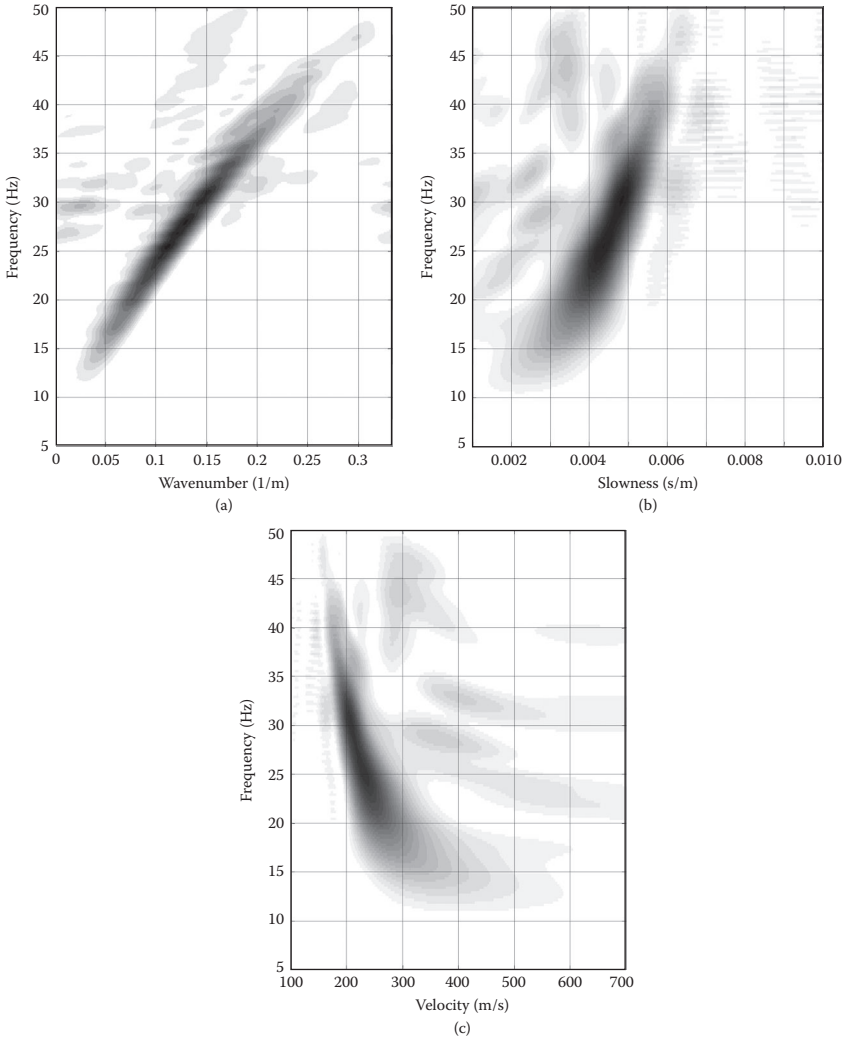


Figure 7.7 Amplitude spectra in different domains of the experimental data of Figure 7.1b: (a) frequency–wavenumber domain; (b) frequency–slowness domain; (c) frequency–velocity domain.

The inversion has been performed using a local search approach based on a weighted and damped least squares algorithm (Herrmann 1994). A comparison among different inversion schemes will be presented in the next section on a different experimental dataset.

Inversion of surface wave data is typically performed with a priori assumptions on some model parameters in order to reduce the number of

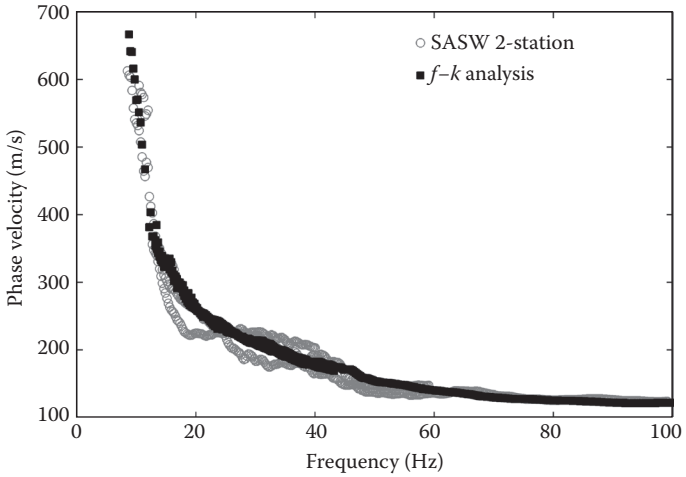


Figure 7.8 Comparison of experimental dispersion curves obtained from the same dataset (see Figure 7.1) with different processing techniques. The squares represent the results obtained with transform-based multistation methods. The other data points are the ones of Figure 7.3, obtained with the two-station method.

unknown parameters. This is suggested in order to mitigate the consequences of ill-posedness of inverse problems. Indeed a large number of unknowns would lead to relevant problems with solution nonuniqueness (i.e., several sets of parameters can be equivalent in terms of comparison between the estimated response of the model and the available experimental information). Moreover, some of the model parameters have little influence on the dispersion curve, and their estimation in the inverse problem would be unreliable because of the limited sensitivity.

In a layered medium with linear elastic homogeneous layers, there are four model parameters for each layer: thickness, density, and two elastic constants. As for the latter, the shear wave velocity (which is directly linked to shear modulus and density) and Poisson ratio are typically used as reference parameters.

Most often the inverse problem is resolved assuming a priori values for density and Poisson ratio of each layer. Material density can be estimated a priori with a sufficient level of confidence with a basic knowledge of the local geology.

The minor influence of the Poisson ratio on surface wave propagation in a homogeneous elastic half-space (Richart et al. 1970) could initially justify the choice of assuming an a priori value for this parameter when interpreting experimental data.

A parametric study was performed by Nazarian (1984) to assess the relative influence of model parameters on the dispersion curve. Considering the

similarity among numerical curves obtained for different values of density and of Poisson ratio over a wide frequency range, he concluded that the influence of density and Poisson ratio was negligible compared to that of layer thickness and shear wave velocity. But in that parametric study, only simultaneous variations of the Poisson ratio for all the layers were considered. Different values for different layers can produce variations in the overall shape of the dispersion curve that may significantly influence the solution of the inverse problem.

In particular, a relevant effect may be associated with the water table position at the site. Indeed, for a dry soil, the range of variation of the Poisson ratio is about 0.1 to 0.3, whereas for a saturated porous medium, the apparent Poisson ratio is very close to 0.5 because in terms of wave propagation, an undrained behavior is expected. Substantial errors can be driven by the wrong hypothesis about the water table position, as will be shown in the following discussion.

Information on water table position at the Saluggia site can be inferred from a P-wave refraction survey that has been carried out together with surface wave tests. In fact, active-source surface wave tests and P-wave seismic refraction share a very similar testing setup. For this reason, it is interesting to evaluate the synergies between the two tests (see Section 6.4.4 for further details). P-wave refraction surveys in shallow sediments are very sensitive to the water table position because of the abrupt change in P-wave velocity associated with the passage from unsaturated to saturated conditions. At the Saluggia site, a simple interpretation of P-wave arrival times (assuming a horizontally layered system, which is reasonable for the geological context) shows an interface at a depth of 3 m, which can be assumed to be the position of the water table.

In order to show the importance of the assumption of the water table position, three different inversions of the experimental dispersion curve in Figure 7.5 are compared. The inversions have been performed with the same code, assuming the same starting V_S profile (constant V_S with depth). Number and thicknesses of the layers have been assumed a priori on the basis of a previous interpretation of the same surface wave dataset (Foti 2000).

Inversion #1 has been conducted assuming the correct position of the water table (3 m below the ground surface), with values of density and Poisson ratio of a saturated medium (2100 kg/m^3 and 0.49, respectively) below this level. Inversion #2 (no water table) has been conducted assuming a constant value of density and Poisson ratio with depth (1800 kg/m^3 and 0.2, respectively). Inversion #3 assumes an incorrect position for the water table (7 m below the ground surface).

In terms of shear wave velocity profiles, the final results are reported in Figure 7.9 and are compared to the independent results of a cross-hole test at the same location. The dispersion curves (Figure 7.10) show the perfect

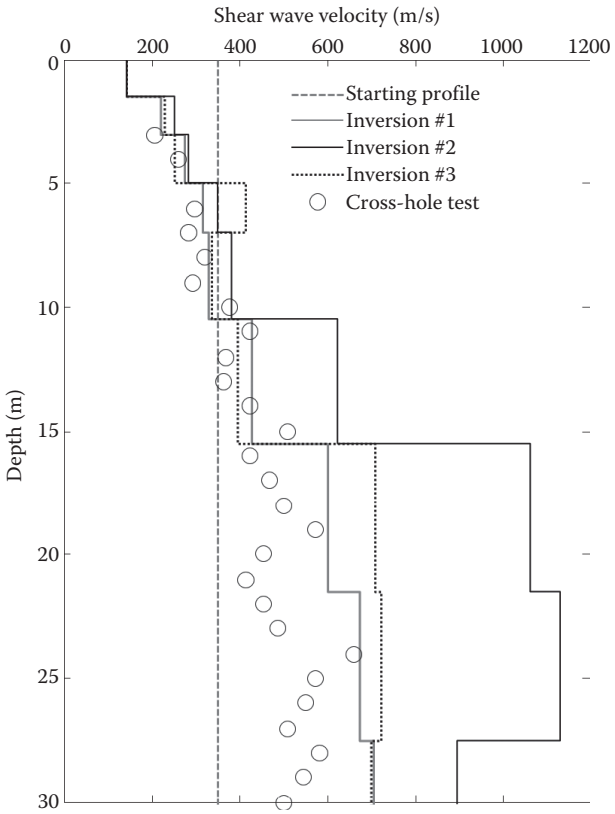


Figure 7.9 Relevance of the hypothesis on water table position on the inversion of surface wave data. Results of the inversion are compared to the shear wave velocity profile obtained with a cross-hole test. (From Foti, S., and C. Strobbia, Some notes on model parameters for surface wave data inversion. In: Proceedings of SAGEEP 2002, Las Vegas, USA, February 10–14, 2002, CD-Rom.)

equivalence among the results of the different inversions and a good match with the experimental dispersion curve. The shear wave velocity profiles show great discrepancies. Inversion #1, with a priori assumptions on the density and the Poisson ratio consistent with available information on the water table position, provides a shear wave velocity profile with a trend of gradual increase with depth that is in agreement with the stratigraphy of the site and with the results of the cross-hole test. The hypothesis of the absence of the water table (Inversion #2) is strongly misleading and results in severe inaccuracies in the final results. The wrong hypothesis on the position of the water table (Inversion #3) leads to an overestimation of the shear wave velocity profile and hence of the soil stiffness at depth. These errors

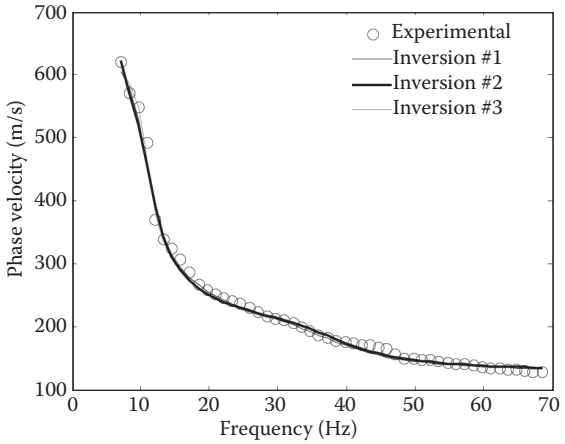


Figure 7.10 Experimental dispersion curve compared to the theoretical dispersion curves for the three profiles in Figure 7.9. (From Foti, S., and C. Strobbia, Some notes on model parameters for surface wave data inversion. In: Proceedings of SAGEEP 2002, Las Vegas, USA, February 10–14, 2002, CD-Rom.)

are driven by the necessity of following the trend of increasing Rayleigh wave velocity for decreasing frequency. This trend is partially generated by the higher Poisson ratio, but it is interpreted only as a difference in shear wave velocity with depth for Inversion #2 and partially for Inversion #3.

This example shows the importance of a proper selection of a priori values for parameters that are not considered as unknown in the inversion process. Cautious and scrupulous judgment by the operator is required in this instance. All the available information on the site has to be taken into account in the selection.

7.2 COMPARISON AMONG INVERSION STRATEGIES

A comparative analysis of inversion strategies is presented in this section. The test site is part of a wide alluvial fan in the Italian Alps. In this case, experimental data for surface wave analysis have been collected using an active-source multistation setup and a circular array for microtremors. The combination of active-source and passive data is often adopted to extend the characterization to significant depths when heavy seismic sources such as truck-mounted Vibroseis are not available.

7.2.1 Experimental dataset

The active-source dataset has been collected using an 8 kg sledgehammer as the seismic source and 48 4.5 Hz vertical geophones. The spacing between

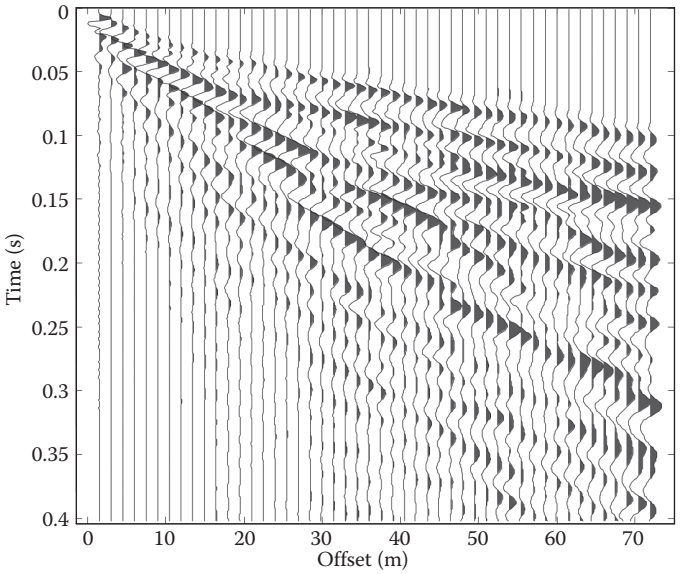


Figure 7.11 Experimental data for active-source setup (48 receivers with 1.5 m spacing and 6 kg sledgehammer).

geophones is 1.5 m for a total array length of 70.5 m. A total of 30 shot gathers have been collected for subsequent vertical stacking and analysis of experimental uncertainty. The resulting stacked shot gather is reported in Figure 7.11, and it shows surface waves preceded by large amplitude body waves.

The corresponding frequency–wavenumber spectrum is reported in Figure 7.12. From the location of spectral maxima in Figure 7.12, it is possible to obtain the experimental dispersion curve in the frequency range 10–40 Hz.

Passive data have been collected with 2 Hz vertical geophones deployed in a circular array of 75 m diameter. The spectral analysis has been performed with the frequency domain beamformer method as implemented by Zywicki (1999). Some examples of contour plots of the spectral power density are reported in Figure 7.13. Each spectrum is plotted as a function of wavenumbers in two orthogonal directions for a given frequency. The origin of the axes coincides with the center of the acquisition array. The position of the energy peak identifies the direction of the arrival of the main perturbation with respect to the center of the array and allows for the identification of the wavenumber associated with the specific frequency, which can be used to evaluate the velocity of propagation. A circle is plotted on top of each spectrum: its radius is the wavenumber corresponding to the identified phase

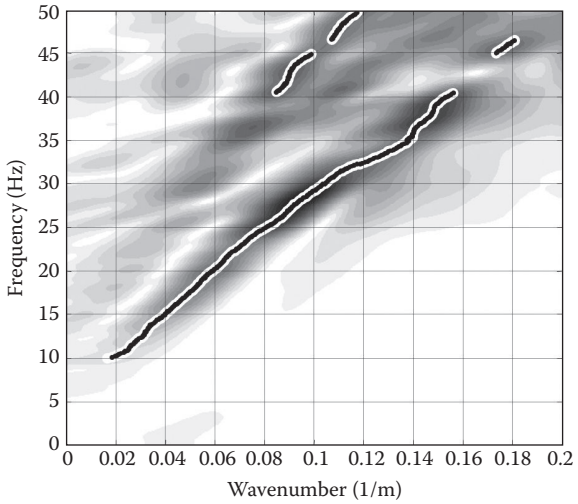


Figure 7.12 Analysis in the *f–k* domain of experimental data of Figure 7.11, amplitude of the frequency–wavenumber spectrum and its absolute maxima.

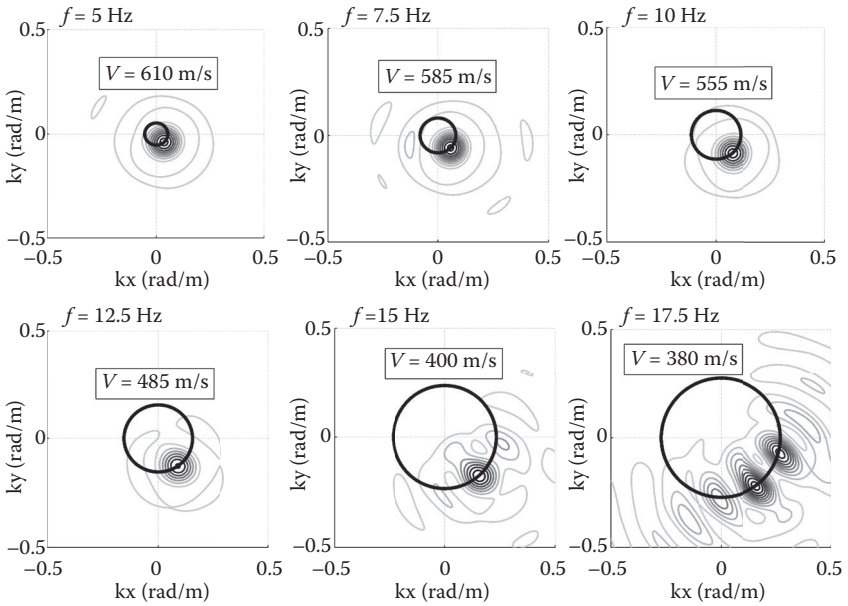


Figure 7.13 Analysis of passive data in the frequency–wavenumber domain, examples of amplitude of the spectrum for different frequencies. The circles represent the dominant wavenumber from which the phase velocity for the specific frequency is evaluated. (From Foti, S. et al., *Riv Ital Geotecnica*, 41(2), 39–47, 2007.)

velocity at the specific frequency value. The identification of phase velocities is repeated for all the frequencies in the range of interest in order to evaluate the experimental dispersion curve for Rayleigh waves.

At this site, waves are arriving from a specific direction that can be identified from the location of the maxima, which is consistent in the different panels. The main source of microtremors was likely associated with construction activities in an area a few hundreds of meters away from the site. In such conditions, special care should be exerted in the choice of the processing method, avoiding the methods that assume homogeneous distribution of the sources in space around the array (specifically spatial autocorrelation and its modification). This example also shows the great potential of this technique to identify the sources of vibrations, which can be useful for studies related to mitigation of ground vibrations caused by industrial or other human activities (Comina and Foti 2007). The source identification may also be relevant for forensic applications.

In order to evaluate the influence of uncertainties in the experimental data, independent processing of datasets collected with the same configuration has been performed for active-source and passive methods. This procedure allows for an evaluation of uncertainty related to background noise but other effects—such as errors in the geometry—cannot be assessed, so that only a partial estimate of the overall uncertainty is obtained. The whole set of experimental dispersion curves obtained for active-source and passive tests is reported in Figure 7.14a; Figure 7.14b reports the corresponding mean values and standard deviations. The uncertainties in active and passive tests are very different. This is even more evident in Figure 7.15, which shows the coefficient of variation of the experimental data, defined as the ratio between standard deviation and mean value of the phase velocity at each frequency. The values of the coefficient of variation obtained from active data are in line with those reported in other studies (Marosi and Hiltunen 2004a; Lai et al. 2005); therefore, these values can be used as a reference when a direct estimate is not feasible.

In the following example, the experimental data gathered with the combination of active and passive testing setup are used for a comparative analysis of inversion approaches. The interpretation assumes that the estimated experimental dispersion curve corresponds to the fundamental Rayleigh mode. This hypothesis appears reasonable on the basis of local geology because a gradual increase of stiffness with depth is expected considering the typical formation process of alluvial fans in the Alps and given the mix-grained nature of the material. Moreover, the trend observed in the experimental dispersion curve is consistent with a typical fundamental mode at sites with gradual increases of stiffness with depth.

An inspection of the experimental dispersion curve can provide some useful hints for the solution of the inverse problem. Taking into account maximum and minimum available wavelength, it is possible to have an indication of the investigation depth and the expected resolution close to the ground surface.

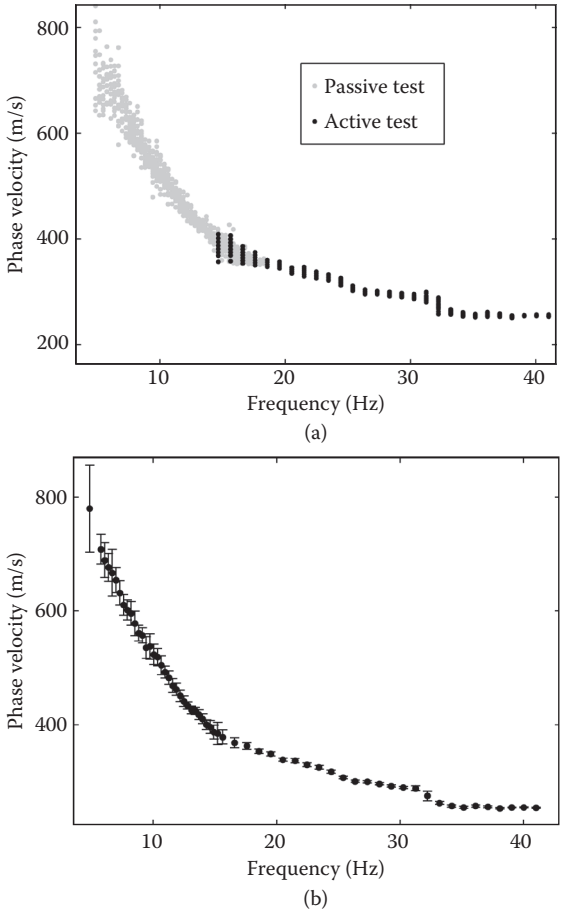


Figure 7.14 Experimental dispersion curve from the combination of active-source and passive data: (a) ensemble of data points for several repetitions of the test with the same geometrical setup; (b) average dispersion curve and its associated standard deviation evaluated with the data in (a). (From Foti, S. et al., *Riv Ital Geotecnica*, 41(2), 39–47, 2007).

The investigated depth is about half of the maximum available wavelength. Considering the data point at the lowest available frequency

$$\lambda_{\max} = \frac{V_R}{f} = \frac{780}{5} = 155\,\text{m} \tag{7.1}$$

Hence, the investigated depth is about 80 m. Layers below this depth would not be sufficiently constrained by the experimental data, and it is inappropriate to include them in the inversion.

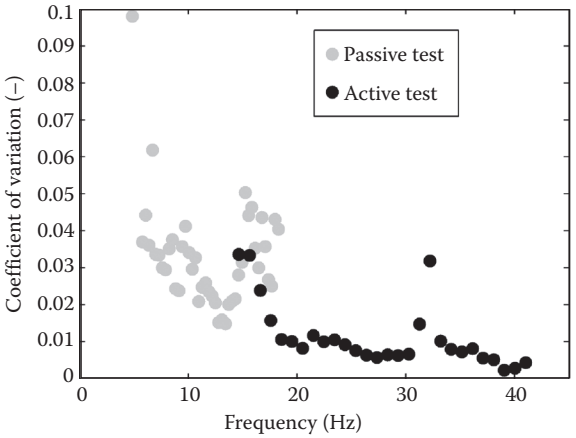


Figure 7.15 Coefficient of variation (standard deviation over average values) for the experimental dispersion curve of Figure 7.14.

As for the resolution at shallow depth, the thickness of the first layer that can be adequately reconstructed in the model is about half the minimum wavelength. The latter is estimated on the basis of the high-frequency components

$$\lambda_{\min} = \frac{V_R}{f} = \frac{250}{41.5} = 6\,\text{m} \tag{7.2}$$

7.2.2 Empirical inversion

A preliminary estimate of the shear wave velocity profile in normally dispersive sites with stiffness gradually increasing with depth may be obtained with the mapping of the points of the dispersion curve into approximate values of shear wave velocity with depth (Section 6.3).

Each point of the experimental dispersion curve is transformed into a point of the pseudo–shear wave velocity profile by associating a shear wave velocity slightly higher than the phase velocity of the Rayleigh wave at that frequency (e.g., V_S = 1.1 V_R) to a depth equal to a fraction of the wavelength (e.g., 1/2.5). This procedure is derived from the observation of the wave propagation parameters in a homogeneous linear elastic half-space. Application to the experimental dispersion curve of Figure 7.14b gives the pseudo–shear wave velocity trend of Figure 7.16a.

This approach cannot be considered a substitute of a proper inversion process, but it can be useful for estimating a first tentative profile to be used as a starting profile (Figure 7.16b) in iterative procedures for automatic inversion or for trial and error procedures.

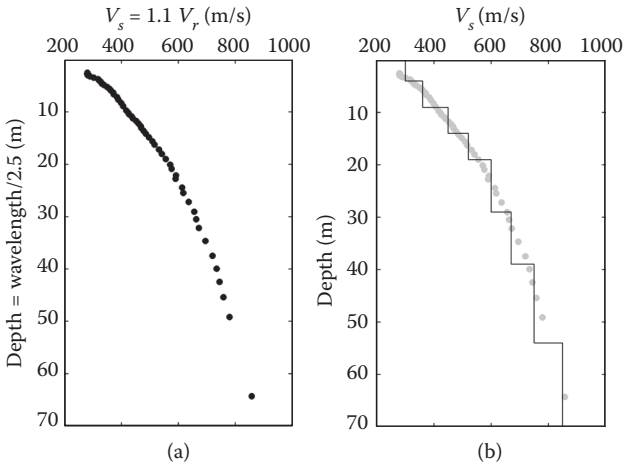


Figure 7.16 Estimate of the shear wave velocity profile with direct mapping of the point of the experimental dispersion curve: (a) pseudo–shear wave velocity profile; (b) construction of a first tentative profile for subsequent solution of the inverse problem.

7.2.3 Deterministic approach (least squares)

The deterministic inversion of the mean values of the experimental dispersion curve based on a weighted damped least square algorithm (Herrmann 1994) is reported in Figure 7.17. The profile of Figure 7.16b has been used as a starting model for the iterative procedure. The obtained shear wave velocity profile is compared to the results of down-hole tests at two sites in the same alluvial fan. The actual distance between each of the three sites is about 1 km so a direct comparison is not appropriate, but considering the nature of the sediments, the agreement appears to be meaningful anyway.

In order to obtain the standard deviation of the final shear wave velocity profile, the standard deviation of the Rayleigh wave phase velocities (Figure 7.17) can be incorporated in the inversion process using a linearization in the neighborhood of the final solution (Tarantola 2005). The corresponding error bars on the velocity profile of Figure 7.18 have been obtained using the procedure described in Section 6.5.3.

The low experimental uncertainty of active-source data in the high-frequency range (Figure 7.17) leads to very low uncertainties in the velocity estimates for shallow layers (Figure 7.18). For deeper layers, the higher uncertainty of passive data leads to higher uncertainties in the shear wave velocity. The coefficient of variation of shear wave velocity ranges from less than 1% for shallow layers to about 6% for deeper layers, showing that the inversion process is not affected by error magnification.

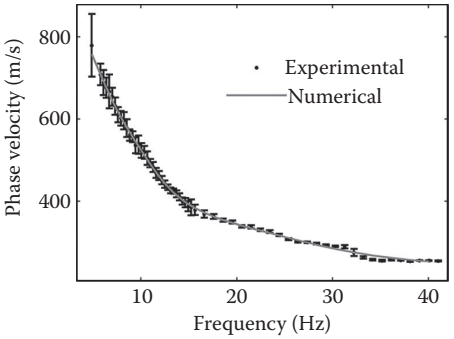


Figure 7.17 Local search inversion with damped least square algorithm—experimental dispersion curve and its associated standard deviation compared to the theoretical dispersion curve for the best-fitting shear wave velocity profile. (From Foti, S. et al., *Riv Ital Geotecnica*, 41(2), 39–47, 2007.)

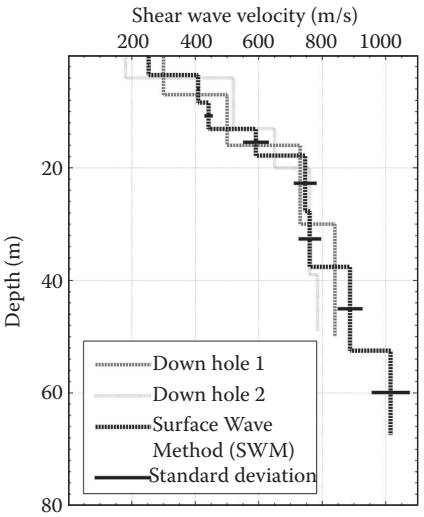


Figure 7.18 Local search inversion with damped least square algorithm—best-fitting shear wave velocity profile and its associated standard deviation compared to results of down-hole tests at locations nearby. (From Foti, S. et al., *Riv Ital Geotecnica*, 41(2), 39–47, 2007.)

7.2.4 Stochastic approach (Monte Carlo)

Deterministic inversions with local search methods provide a solution that may represent a local minimum of the misfit function (typically a norm of the difference between experimental and theoretical dispersion curves). The local minimum in which the solution is trapped depends also

on the starting profile for the iterative procedure. In order to mitigate the risk of getting trapped into a local minimum, global search methods can be applied.

Moreover, a stochastic approach can provide an assessment of the solution's nonuniqueness. Indeed, for mathematically ill-posed problems, such as inverse problems, the solution is not unique, and several solutions can be considered equivalent because they provide the same or similar agreement with experimental data. In this respect, it is also necessary to account for uncertainties in experimental data when comparing them to the theoretical dispersion curve of possible solutions.

Experimental data for the La Salle site have been also interpreted with a global search method based on a Monte Carlo approach, using the code by Socco and Boiero (2008). It allows a limited population size to be used because it applies scale properties of modal curves (Socco and Strobbia 2004; Maraschini et al. 2011) to improve the efficiency in sampling the model parameter space. Moreover, the use of scale properties allows model parameters to be outside the boundaries initially selected for the generation of the population (Socco and Boiero 2008). The application of a statistical test (one-tailed Fisher test) leads to the selection of a set of acceptable shear wave velocity profiles according to data quality and model parameterization (Socco and Boiero 2008). Selected V_S velocity profiles represent the final result of the inversion for a chosen level of confidence. The results reported in Figure 7.19 have been obtained with a population of 10^5 models. Well-resolved parameters produce a small range of variations, while poorly constrained parameters can assume virtually any value within a wide range.

Figure 7.20 shows all the theoretical dispersion curves for the selected profiles of Figure 7.19 (compared to the experimental one) and also shows the equivalence of these profiles. The hierarchical relative misfit representation adopted for the equivalent profiles (Figure 7.19) is such that the darkest color corresponds to the profile having the lowest misfit. The deterministic inversion of the previous section and its associated standard deviation are also reported, for comparison, in Figure 7.19.

High resolution and low uncertainty are attained for shallow layers, although a much higher uncertainty from nonuniqueness is obtained for deeper layers. This is due to the fact that the resolution of surface waves inevitably decreases with depth and to the higher experimental uncertainty in the passive tests (Figure 7.15). The uncertainty in the first layer looks negligible, both for its velocity and its thickness, thanks to the large amount of data available to constrain these parameters in the inverse problem solution. It has to be stressed that the obtained parameters correspond to the average laterally homogeneous model. The model errors, in particular those due to the lateral variations, are not estimated.

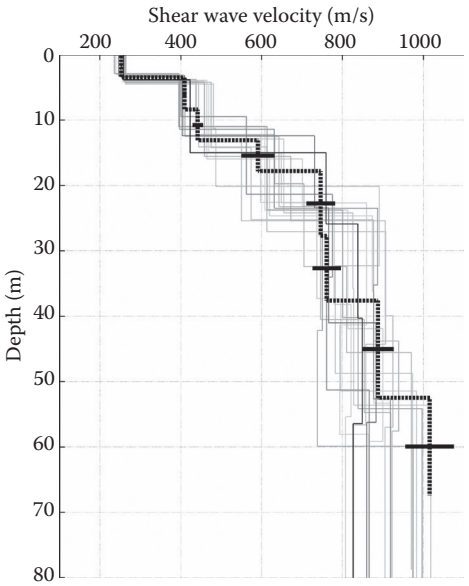


Figure 7.19 Global search inversion with Monte Carlo approach—best-fitting shear wave velocity profiles selected with a statistical test compared to the solution of Figure 7.18 obtained with the local search method (LSM). The profiles are plotted in a grayscale color range according to their associated misfit (the darker the line, the lower the misfit). (From Foti, S. et al., *Riv Ital Geotecnica*, 41(2), 39–47, 2007.)

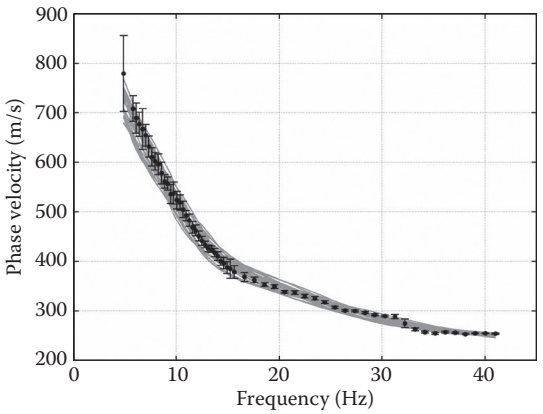


Figure 7.20 Global search inversion with Monte Carlo approach—experimental dispersion curve and its associated standard deviation compared to the theoretical dispersion curve for the best-fitting shear wave velocity profiles of Figure 7.19 selected with a statistical test. (From Foti, S. et al., *Riv Ital Geotecnica*, 41(2), 39–47, 2007.)

It can be interesting to assess the consequences of solution nonuniqueness in terms of the specific application for which site characterization is performed. In this respect, Figure 7.21 shows the response spectra obtained for the same strong motion record by assuming each one of the shear wave velocity models of Figure 7.19. Clearly, the differences are quite limited. If no a priori information is used to constrain the results, errors in the identification of a single parameter can be very high. Nevertheless, the consequences of such uncertainties are very limited for seismic site response studies, at least where a 1D approach is deemed to be reasonable. Indeed, if shear wave velocity profiles are equivalent with respect to Rayleigh wave dispersion, they are also equivalent with respect to seismic ground response. Different profiles that represent the global response of the site in terms of Rayleigh wave propagation are also representative of the seismic response of the soil deposit.

Another relevant issue is related to the estimation of $V_{S,30}$ (i.e., the average velocity of the shallowest 30 m), which is often adopted in seismic studies and in seismic codes for subsoil classification (Borcherdt 1994). The estimate of $V_{S,30}$ for each of the equivalent shear wave velocity profiles in Figure 7.19 is reported in Figure 7.22. Because $V_{S,30}$ is an average parameter, the influence of solution nonuniqueness on its estimate is limited, and the range of variation of the different estimates is very narrow.

A preliminary estimate of $V_{S,30}$ can be obtained directly from the experimental dispersion curve. The idea that the dispersion curve offers a consistent and simplified way of estimating the $V_{S,30}$ has been considered by Brown et al. (2000), who proposed the Rayleigh wave phase velocity at a

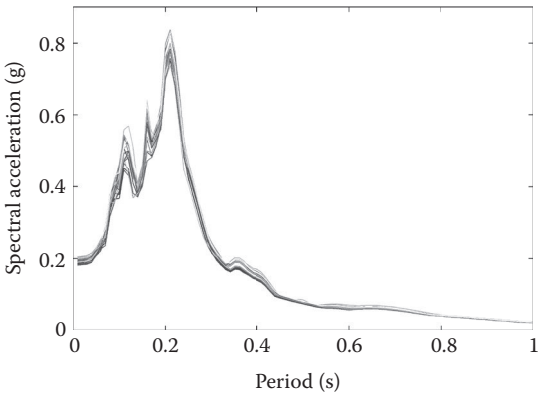


Figure 7.21 Simulation of 1D seismic ground response for the group of equivalent shear wave velocity profiles of Figure 7.19. (From Foti, S. et al., *Soil Dynam Earthquake Eng*, 29(6), 982–993, 2009.)

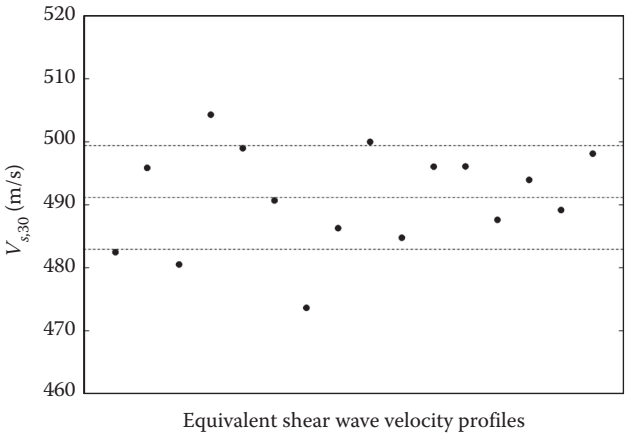


Figure 7.22 $V_{S,30}$ evaluated for each of the equivalent shear wave velocity profiles of Figure 7.20.

wavelength of 36 m as a direct estimate of $V_{S,30}$. Such an approach cannot be considered as an alternative to the solution of the inversion process because this oversimplification can lead to large errors, especially for complex stratigraphical situations. Nevertheless it can provide a useful reference for preliminary assessments. For the experimental dispersion curve of Figure 7.14, this estimate would be equal to 450 m/s, which is about 10% less than the values estimated with the shear wave velocity profiles obtained from the inversion (Figure 7.22).

7.2.5 Deterministic approach with vertically heterogeneous medium

The inversion results presented in the previous sections assumed a layered model for the investigated subsoil. This is the typical approach in surface wave testing. This approach provides the most reliable results where geological conditions are such that the expected stratigraphy actually consists of alternate layers of materials, internally homogenous and having different seismic properties. In other situations (e.g., homogenous deposits of coarse grained materials in which the variation of stiffness with depth is mainly due to the effect of increasing confining pressure), a gradual and continuous increase of shear wave velocity is expected. In such situations, the adoption of a layered model leads to an approximation of actual field conditions, and the gradual increase is represented as a step-like increase. The interpretation of surface wave data can be performed with a more appropriate model allowing for continuous variations of seismic properties (see Section 2.4).

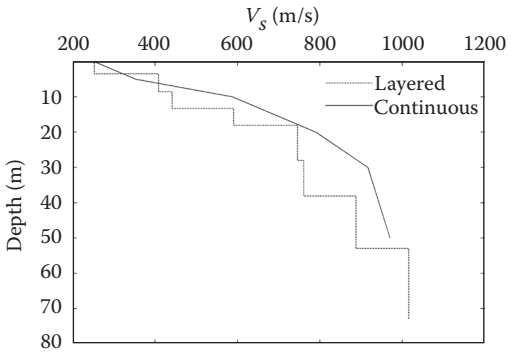


Figure 7.23 Comparison of shear wave velocity profiles obtained with local search methods using different reference models—smooth vertical heterogeneous elastic model versus stack of linear elastic layers.

As mentioned before, the La Salle site is part of a large alluvial fan, which is mainly composed of chaotic material. The inversion of the dispersion curve of Figure 7.14b, in terms of smooth variations, has been performed with the code developed by Rix and Lai (2013). The results are shown in Figure 7.23 and are compared to the results obtained with the classical layered medium approach (Figure 7.18). For the very shallow portion of the profile, which shows a marked jump of shear wave velocity likely associated with the superficial layer of residual and vegetated soil, the interpretation in terms of layered medium is likely more appropriate.

7.3 EXAMPLES FOR DETERMINING V_S AND D_S PROFILES

A crucial aspect in site characterization for seismic applications is related to the quantification of material damping. Indeed, internal energy dissipation in hysteretic cycles plays a major role in the definition of seismic ground response. Laboratory tests provide an evaluation over a wide strain range, but they are performed on a limited number of samples; hence, they represent a local estimate not representative of the behavior of the whole deposit. *In situ* seismic tests can in principle be used to characterize large volumes, but their implementation faces difficulties related to the necessity of separating geometric attenuation and material attenuation in wave propagation phenomena. Surface wave data can be interpreted to provide an estimate of material damping. The methods for the estimation of the experimental attenuation curve have been discussed in Chapter 5, whereas the inversion procedures to get the material damping profile are reported in Chapter 6. The application of the whole procedure to experimental data is reported in the following two examples.

7.3.1 Memphis

This testing site is located on Mud Island near downtown Memphis, Tennessee. Mud Island was originally formed by dredge spoil taken from the Mississippi River. The area currently forms a peninsula located on the northwestern edge of downtown Memphis. The water table is 8 m below the ground level. Surface wave tests were performed with a multistation array and interpreted with the transfer function approach presented in Section 5.3.

The wave train was generated by a vertically oscillating, electrodynamical shaker operating in swept-sine mode. The frequency range used in the field test was from 3.75 to 100 Hz. Rayleigh wave particle motion was recorded by vertical accelerometers having a flat frequency response from 0.10 to 300 Hz. The accelerometers were placed in a linear array at the following offsets from the source: 2.44, 3.05, 3.66, 4.57, 5.49, 6.71, 8.54, 10.37, 12.80, 15.24, 18.29, 21.34, 24.39, 28.96, and 33.54 m. The acceleration of the armature of the electrodynamical shaker was also measured with a piezoelectric accelerometer to allow the input force to be calculated. The experimental transfer function was obtained at each receiver location from an average in the frequency domain of 10 measurements to reduce the variance of the measured spectral quantities.

The complex-valued transfer function is used for estimating the complex wavenumber through a regression with the analytical expression of Equation 5.10. An example of the fitting of the regression process is reported in terms of phase and amplitude of the transfer function in Figure 7.24. The process is repeated over the frequency range scanned with the harmonic source in order to simultaneously obtain the experimental dispersion and attenuation curves (Figure 7.25). The inverse problem aimed at estimating

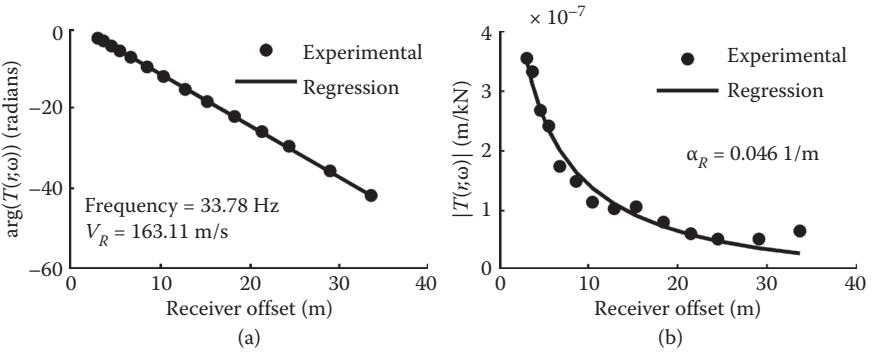


Figure 7.24 Regression of the complex-valued experimental transfer function for a sample frequency: (a) plot of the phase and (b) plot of the amplitude as a function of offset. (From Lai, C. G. et al., *Soil Dynam Earthquake Eng*, 22(9–12), 923–930, 2002.)

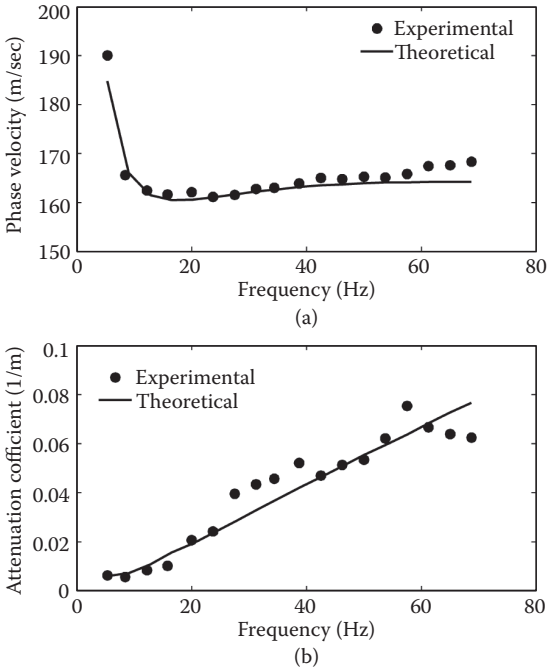


Figure 7.25 Comparison between experimental and theoretical dispersion (a) and attenuation (b) curves (fundamental mode). (From Lai, C. G. et al., *Soil Dynam Earthquake Eng*, 22(9–12), 923–930, 2002.)

the shear wave velocity and damping ratio profiles has been solved with a damped least square approach with coupled and simultaneous inversion (see Section 6.4.3.4). The results are reported in Figure 7.26, where the shear wave velocity profile is also compared to results of a down-hole survey performed with the seismic cone (Mayne 2000).

7.3.2 Pisa

The site of the Leaning Tower of Pisa has been extensively characterized over the years for studies on the stability of the tower and for planning the remediation works for its safeguard. The local stratigraphy, from top down, is as follows. The first formation is predominantly slightly clayey and sandy silt with lenses of sand and clay. It extends down to a depth of around 10 m. The underlying clayey formation has a thickness of approximately 30 m. In particular, the zone between 10 and 20 m from the ground surface is a very uniform deposit of soft clays that was the main cause of the differential settlement during the construction of the tower. The water table is shallow (2 to 3 m below the ground surface) and is subject to marked seasonal fluctuations.

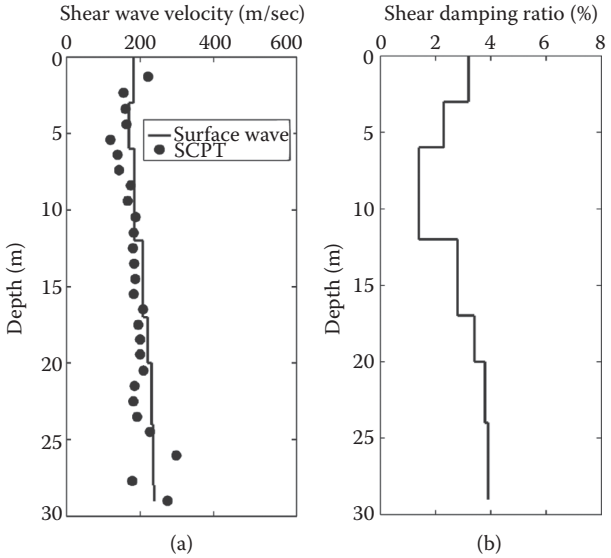


Figure 7.26 Shear wave velocity (a) and shear damping ratio (b) profiles obtained with simultaneous inversion of experimental dispersion and attenuation curves of Figure 7.25. The shear wave velocity profile is compared to the results of a down-hole test performed with the seismic cone penetration test (SCPT). (From Lai, C. G. et al., *Soil Dynam Earthquake Eng*, 22(9–12), 923–930, 2002.)

The field measurements for surface wave testing were conducted using a 24-channel seismograph and vertical geophones having a natural frequency of 4.5 Hz. Two different seismic impact sources were used to generate energy over a broad frequency range: a weight-drop system (130 kg released under self-weight from a height of 3 m) and an 8 kg sledgehammer for long and short arrays, respectively. The necessity of two different testing configurations is due to the trade-off between the length of the array and the frequency band. The test location is shown in Figure 7.27, in which it is also shown the location of a cross-hole test performed in the vicinity of the Leaning Tower. The lateral homogeneity of the soil deposit across the site allows a meaningful comparison of the results.

Because equal receivers are used at each location, their frequency response is considered equivalent, and the experimental transfer function is evaluated using the deconvolution procedure reported in Section 5.3. An example of experimental transfer function and its regression for obtaining the complex wavenumber is reported in Figure 5.6. By repeating the regression process over the usable frequency range, the experimental dispersion and attenuation curves are obtained (Figure 7.28). The inverse problem has been solved using a damped least squares approach to simultaneously obtain the shear

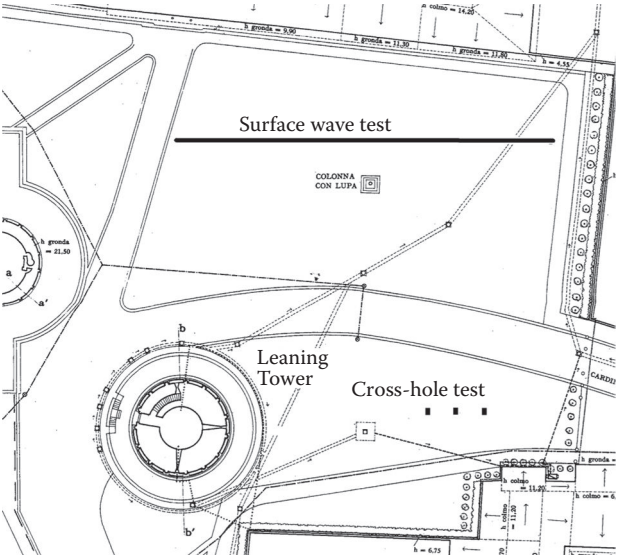


Figure 7.27 Location of surface wave test and cross-hole test at the Leaning Tower of Pisa site.

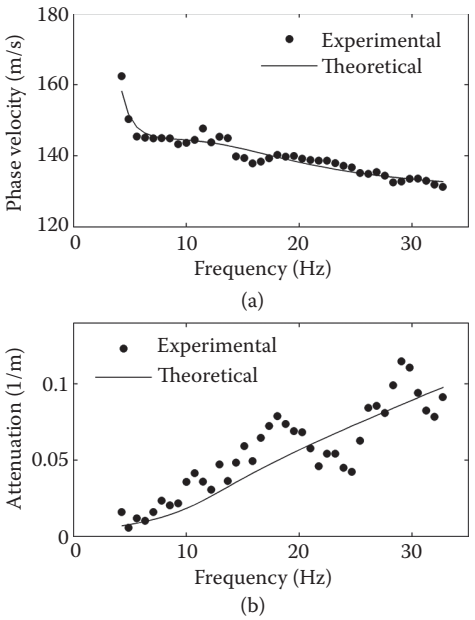


Figure 7.28 Comparison between experimental and theoretical dispersion and attenuation curves (fundamental mode). (From Foti, S., *Geotechnique*, 53, 455–461, 2003.)

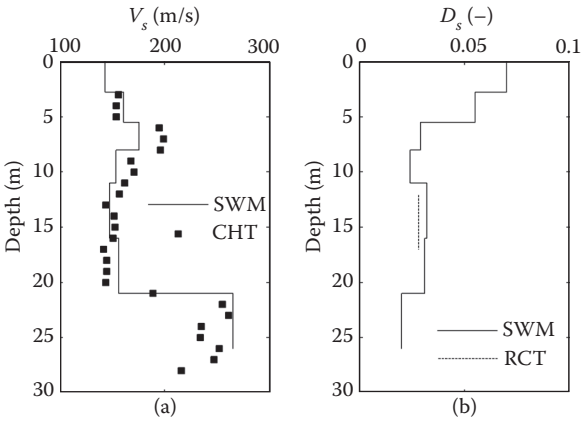


Figure 7.29 Shear wave velocity (a) and shear damping ratio (b) profiles from surface wave method (SWM) obtained with simultaneous inversion of experimental dispersion and attenuation curve of Figure 7.28. The shear wave velocity profile is compared to the results of the cross-hole test (CHT); the damping ratio profile is compared to laboratory data on high-quality samples obtained with the resonant column device (RCT). (From Foti, S. *Geotechnique*, 53, 455–461, 2003.)

wave velocity and damping ratio profiles (Herrmann 1994). The results are reported in Figure 7.29, and they are compared to the other independent experimental data available for the site. In particular, the shear wave velocity profile is compared to the results of the cross-hole test, whereas the damping ratio for the upper clay layer is compared to results obtained in the laboratory with the resonant column device. This comparison shows the reliability of the shear damping profile—at least for the results related to deeper layers. However, compared with usual values reported in the literature, relatively high values are obtained for the uppermost layers. These discrepancies can be partially justified considering that the damping ratio of soils has a frequency dependency more marked in the high-frequency range and that information related to shallow layers is related to high-frequency components of surface waves. Moreover, it has to be considered that energy dissipation phenomena other than geometric and material attenuation are not explicitly accounted for in surface wave testing. In particular, scattering of energy at heterogeneities can introduce another mechanism of attenuation, especially at high frequency; hence, damping ratio values can be somewhat overestimated.

7.4 DEALING WITH HIGHER MODES

One of the main issues with the interpretation of surface wave data is related to the effect of higher modes of propagation and with the difficulties in properly accounting for them in the inversion process. As discussed

in Section 2.4, propagation of surface waves in a vertically heterogeneous medium can be interpreted as the effect of the superposition of free modes of propagation, which are characterized by different dispersion curves. The possibility of identifying individual modes of propagation in experimental data is strongly dependent on the acquisition parameters (specifically on the array length), as discussed in Chapter 3. Often the experimental dispersion curve is indeed an apparent one, arising from the combined effects of different modes of propagation. In such situations, the solution of the inverse problem is not straightforward and requires an adequate strategy to account for higher modes' contribution.

A possible approach is based on the numerical evaluation of an apparent dispersion curve, which takes into account the contribution of higher modes and the layout used in the acquisition of experimental data. The misfit function for the solution of the inverse problem is then defined as a norm of the difference between the experimental dispersion curve and the apparent dispersion curve obtained from the forward model. Two crucial issues are the time-consuming procedure required for the solution of the forward problem and the difficulties in the evaluation of partial derivatives for local search methods.

The following example is based on a different approach in which the misfit function for the solution of the inverse problem is defined in terms of the values of the determinant associated with free surface wave modes (Maraschini et al. 2010). As seen in Section 2.4, the modal curves for free surface waves in a layered (visco-)elastic medium can be obtained by solving an eigenvalue problem, which can be formulated in terms of a propagator matrix representing the dynamic interaction among layers. Examples of such procedures are the transfer matrix method (Thomson 1950; Haskell 1953), the stiffness matrix method (Kausel and Roesset 1981), and the reflection and transmission formulation (Kennet 1974). Wavenumbers that represent the modal solutions can be obtained by imposing the determinant of the matrix to be equal to zero.

If experimental wavenumbers are plugged into the determinant for a given subsoil model, the value of the determinant itself will be different from zero, unless the experimental data are an exact modal dispersion curve for that given model (Figure 7.30). In this sense, the value of the determinant can be considered a measure of the misfit between experimental data and the parameters of propagation in the specific model. With reference to Figure 7.30, the misfit is the sum of the values of the determinant for any specific data point of the experimental dispersion curve. The inverse problem can be solved using this definition of misfit rather than the usual distance between dispersion curves. The advantage of such an approach is twofold. First, it is not necessary to associate a priori the experimental data to the fundamental mode rather than to any of the higher modes. Second, the solution of the forward problem is very light from a computation point of view because time-consuming zero search procedures are avoided. For this reason, the approach is very appealing for global search

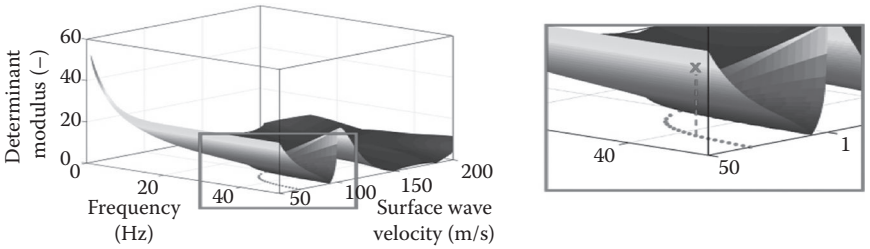


Figure 7.30 Representation of the determinant misfit function. The point indicated with X is the absolute value of the determinant of the synthetic model m* corresponding to an experimental point; this value is the misfit of m* for the single experimental point. (From Maraschini, M., and S. Foti, *Geophys J Int*, 182(3), 1557–1566, 2010.)

methods because very large populations can be used with limited computational efforts. An example of application with a multimodal Monte Carlo procedure (Maraschini and Foti 2010) is reported in the following.

The test has been performed on an unpaved track in which the top layer is an artificially compacted layer of gravels and pebbles. The stratigraphic conditions below this superficial layer report an alluvial–colluvial deposit a few meters thick that lies over weak torbiditic bedrock (interbedded silts and sands). The experimental data have been collected with an array of 24 vertical geophones with a spacing of 0.7 m because of limited area. The short array was considered sufficient as the target of the characterization was very shallow because the seismic bedrock is expected to be close to the ground surface. An 8 kg sledgehammer has been used as a source. The experimental dispersion curve has been obtained with a frequency–wavenumber analysis.

A velocity increase with frequency can be observed in the experimental dispersion curve (Figure 7.31b), suggesting the presence of a velocity inversion in the subsoil model. In this condition, the apparent dispersion curve follows the higher modes in the high-frequency band. On the contrary, the sharp stiffness jump expected at the contact between the alluvial layers and the bedrock could cause a transition to the first higher mode at low frequency as observed in other synthetic and experimental datasets (Foti 2002).

Data were inverted with a Monte Carlo approach considering a population of 10^7 random profiles. The best profiles, which have been selected with a statistical test, are reported in Figure 7.31a. The stiff top layer is coherent with available information on the site. The solutions present a velocity inversion in the shallow portion of the soil profile. The experimental dispersion curve is reported on top of the Haskell–Thomson determinant of the best-fitting profile in Figure 7.31c. Although the inversion process does not require the evaluation of modal dispersion curves, the dispersion curves of the best profiles have also been computed to show their agreement with experimental data (Figure 7.31b).

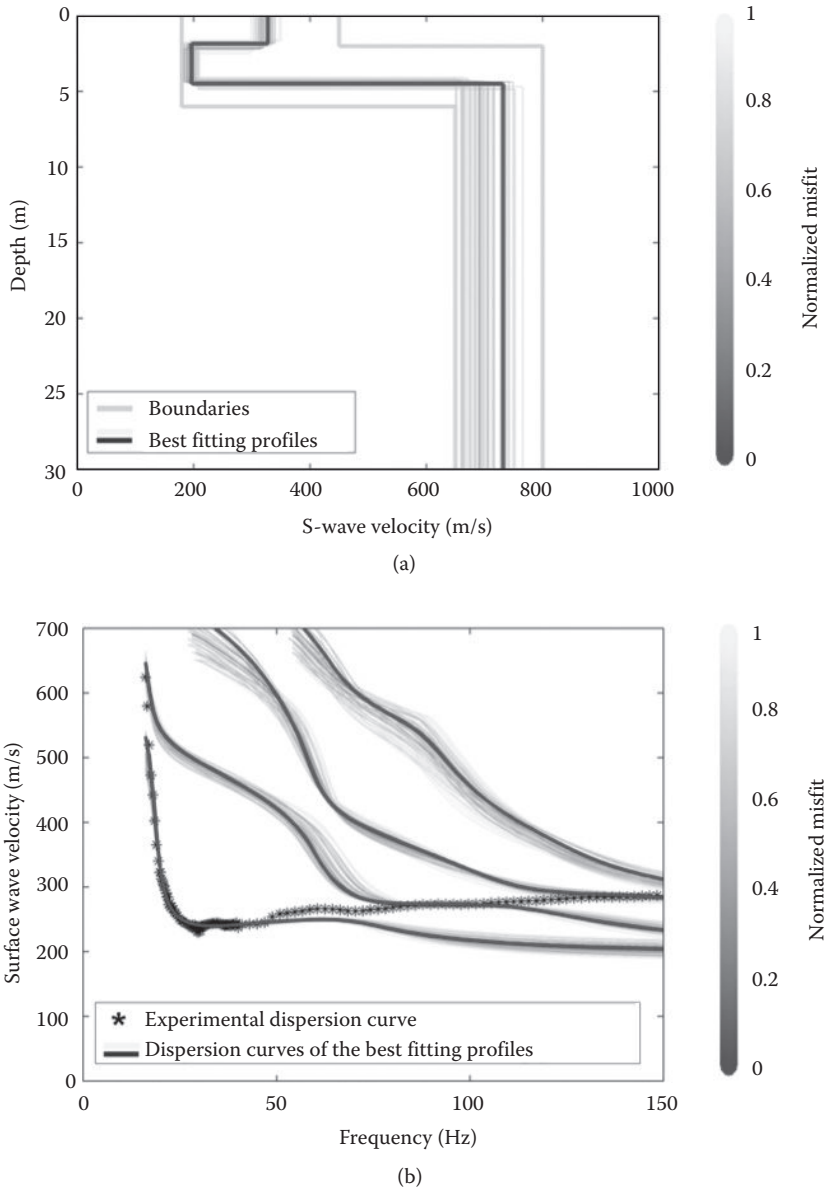
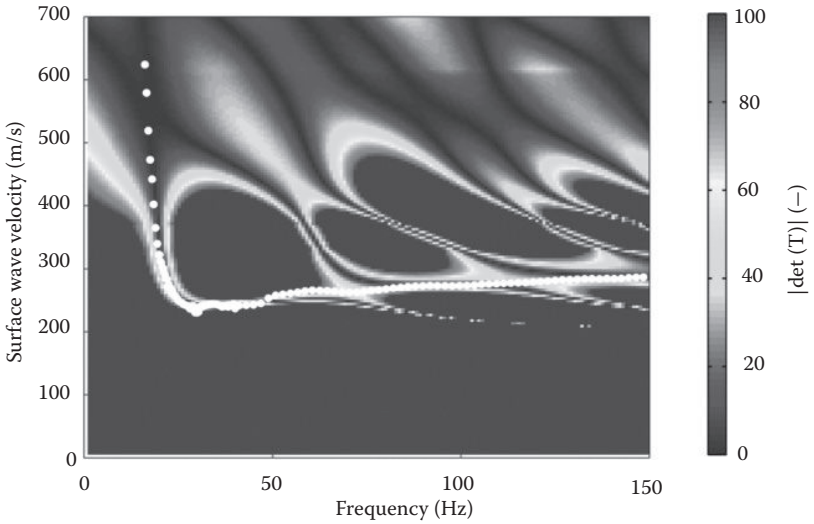


Figure 7.31 Inversion of surface wave data with a Monte Carlo multimodal approach based on the determinant misfit: (a) best-fitting profiles and boundaries for the generation of the population (decreasing misfit from clear to dark); (b) dispersion curves for best models compared with the experimental dispersion curve; same color scale as (a).



(c)

Figure 7.31 (*Continued*) Inversion of surface wave data with a Monte Carlo multimodal approach based on the determinant misfit: (c) absolute value of the Haskell–Thomson matrix determinant for best-fitting model (white dots represent the experimental dispersion curve). (From Maraschini, M., and S. Fot, *Geophys J Int,* 182(3), 1557–1566, 2010.)

This dataset is particularly challenging for a surface wave inversion algorithm because the apparent dispersion curve, composed of a single branch, follows higher modes in the low-frequency band (because of the stiff bedrock) and in the high-frequency band (because of the stiff top layer), with a smooth transition; consequently, the mode numbering identification of the data points is not feasible.

7.5 SURFACE WAVE INVERSION OF SEISMIC REFLECTION DATA

Seismic reflection data often contain high-energy surface waves, usually simply referred to as the "ground-roll". Surface waves have been conventionally considered as coherent noise, superimposing on the useful P-wave reflections signal and filtered out as soon as possible. The conventional reflection acquisition procedures aim at the attenuation of the surface waves in the field, using receiver arrays, sometimes high-frequency geophones, and high-frequency sources—such as vibrators with sweep signals with a rather high minimum frequency.

These acquisition practices can be quite effective in attenuating the surface waves. From the perspective of a surface wave user, these practices damage the surface wave signal, limiting the usable bandwidth and introducing phase distortions. However, if the acquisition setup is not too harsh on the surface waves, reflection data can be particularly interesting. Sources are typically very powerful and provide high signal-to-noise ratio even at far offsets; the offset range usually guarantees high wavenumber resolution and modal separation; the high-fold data can offer an incredible statistical redundancy for the surface wave dispersion extraction.

Some engineering surveys, with 2D acquisition geometry, can contain well-preserved and well-sampled surface waves. The offset ranges, the frequency of receivers, the limited size of the arrays, or the use of single-sensors (or bunched geophones) produce data that are compatible with standard surface wave analysis techniques. The data of modern, large-scale exploration three-dimensional (3D) surveys can also be used, especially with broadband point-receiver acquisition systems.

7.5.1 Application with engineering data in 2D

An example of ground-roll exploitation for the construction of a pseudo–2D shear wave velocity model is reported by Socco et al. (2008, 2009). The reflection dataset has been collected along the two survey lines shown in Figure 7.32. Down-hole tests and specific surface wave tests have also been performed at specific sites as shown in the map and will be used for a comparison. In particular, Site E in Figure 7.32 is the location of the test presented in Section 7.2.

The site is located on a wide triangular alluvial fan. The maximum thickness of the quaternary deposits is expected to be around 200 m, and the fan is mainly composed by alluvial deposits (sand, gravel, stone), polygenic slivers, stones, and blocks. The general geological evolution can be summarized as a succession of alluvial fan deposits having medium to coarse grain size on deposits of glacial environment. The deposit has been preliminary investigated by two boreholes, subsequently used for the down-hole tests. The stratigraphic log to a depth of about 50 m shows the typical chaotic sequences of gravelly soils of alpine alluvial fans, with no marked layering.

Experimental dispersion curves have been extracted from overlapping windows of the seismic reflection seismograms (Figure 7.33). The analysis has been performed with standard transforms in the frequency–wavenumber domain (see Section 4.6.1).

Redundancy of experimental data provided by the several shots of the seismic reflection survey has been used to evaluate reliable average dispersion curves and their associated uncertainty. The experimental dispersion curves for the different spatial windows along the two survey lines are reported in Figure 7.34.

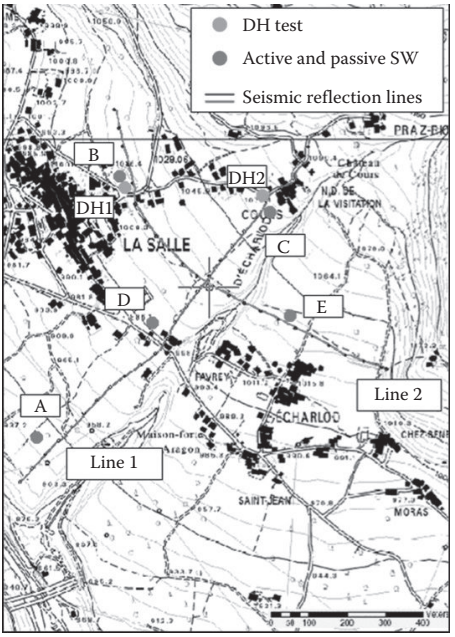


Figure 7.32 Location of seismic surveys performed for the characterization of the alluvial fan in La Salle, Italy. (From Socco, L. V. et al., *Near Surf Geophys*, 6(8), 255–267, 2008.)

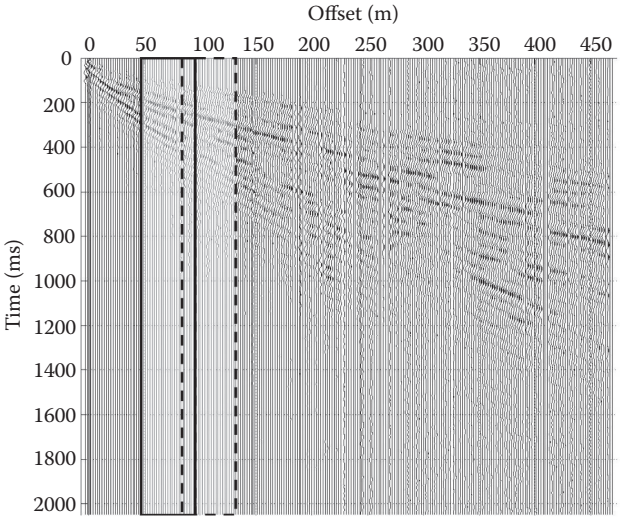


Figure 7.33 Example of experimental dataset from seismic reflection survey with the representation of the sliding window used to perform surface wave analysis.

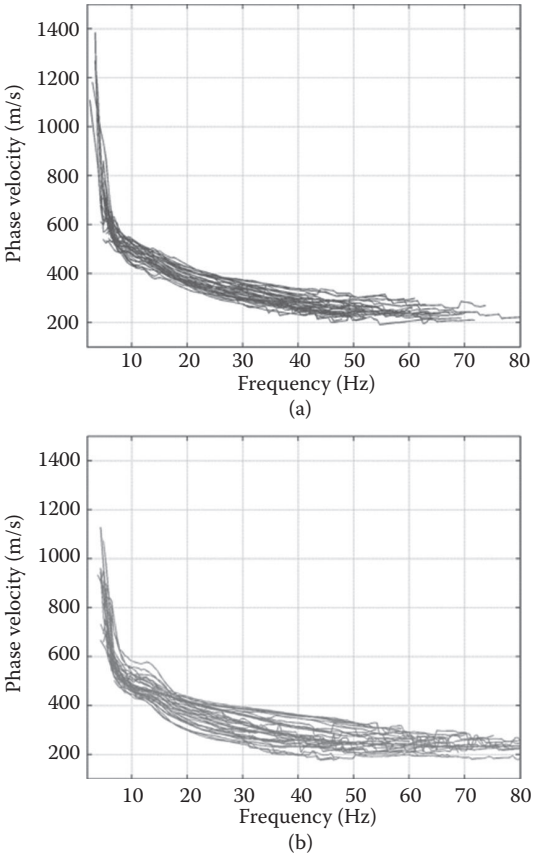


Figure 7.34 Experimental dispersion curves obtained for the different positions of the spatial window along the survey line: (a) Line L1; (b) Line L2. (From Socco, L. V. et al., *Geophysics*, 74, G35–G45, 2009.)

The inversion has been performed with a laterally constrained inversion in which the 1D shear wave velocity profiles associated with the different spatial windows are linked one to each other with lateral constrains which take also into account their distance. The results of the inversion are reported in Figure 7.35. Finally, a comparison of all the geophysical results available for the site is reported in Figure 7.36, where the V_S model obtained with laterally constrained inversion of ground-roll is plotted on top of the results of the seismic reflection survey. In the same figure, the local results obtained from down-hole tests (long arrows) and specific active–passive surface wave tests (short arrows) also are reported. The investigation depth of ground-roll inversion is limited to about 50 m because of the high-frequency geophones used for the reflection survey, which did not allow for the experimental dispersion

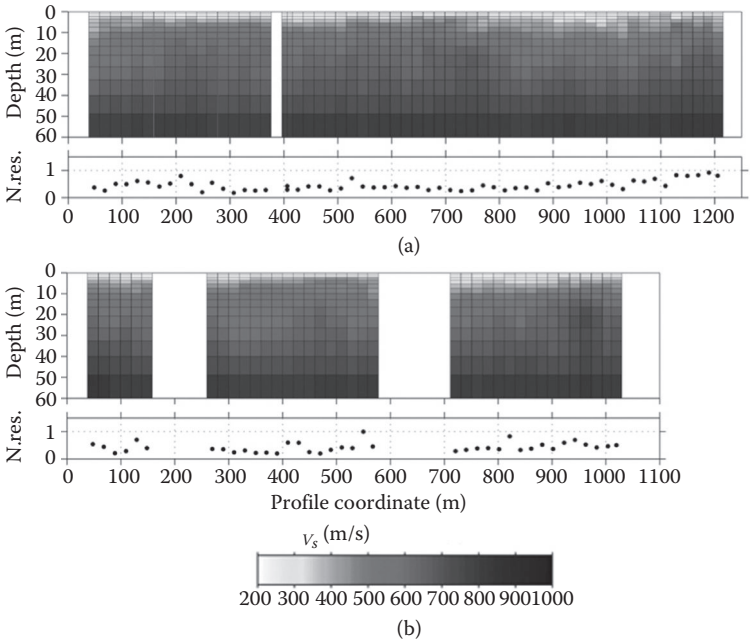


Figure 7.35 Laterally and vertically constrained inversion of the dataset: (a) Line L1; (b) Line L2. The blank sections are related to topographic anomalies where portions of the dataset had to be discarded. The graph below each single profile represents the normalized residual misfit in the inversion. (From Socco, L. V. et al., *Near Surf Geophys*, 6(8), 255–267, 2008.)

curve to be retrieved at low frequency. Deep shear wave velocity profiles obtained with the inversion of combined active and passive surface wave data confirm the deeper reflection as the position of the bedrock below the alluvial fan and allow for the extension of the model at depth (Figure 7.36).

7.5.2 Application with exploration data in 3D

In the exploration industry, acquisition practices are moving toward point-receiver acquisition, in which the wave field is better sampled in space and time. With such high-quality data, surface waves can be analyzed and inverted to characterize the near-surface.

A fine receiver spacing without geophone arrays provides data with non-aliased surface waves, with short wavelengths, and with a dense spatial sampling of the near-surface heterogeneities. Low-frequency sweeps and low-frequency receivers, besides the advantages for the reflection signals, provide the long wavelengths required to reach the large investigation depth in surface wave analysis.

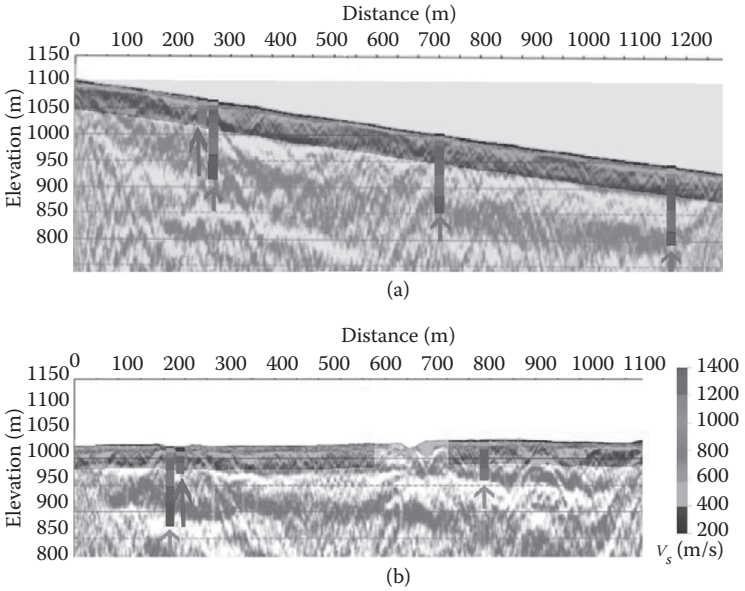


Figure 7.36 Shear wave velocity model obtained from the inversion of surface wave data in ground-roll superimposed to seismic reflection profiles: (a) Line L1 and (b) Line L2. Long arrows and short arrows indicate the results of down-hole tests and active plus passive surface wave tests, respectively. (From Socco, L. V. et al., *Near Surf Geophys*, 6(8), 255–267, 2008.)

For surface wave analysis and inversion, the use of 3D geometries is not a limitation but offers advantages and enables robust processing. With land 3D configurations—for example, cross-spread orthogonal shooting (Strobbia et al. 2010)—surface wave propagation is well sampled in the offset and azimuth domains. A very large statistical redundancy is available for the surface wave analysis. Local supergathers can be formed for the surface wave analysis, collecting traces corresponding to a set of receivers over an area within a defined aperture. The number of shots contributing to a single location can be very large, of the order of thousands; this extreme redundancy of high-fold land data can be exploited to enhance and extract linear events. Transform-based methods can be used to extract the dispersion. In order to process data from 3D geometries, the transforms must be able to handle unevenly spaced arrays and 2D arrays.

For source and receiver lines, the dispersion properties can be plotted as pseudosections for each individual mode. In this representation, the position along the line is the horizontal axis; the vertical axis is the wavelength (related to the investigation depth); and the color scale represents the local phase velocity. An example of continuous profiling along a 10 km receiver line is shown in Figure 7.37.

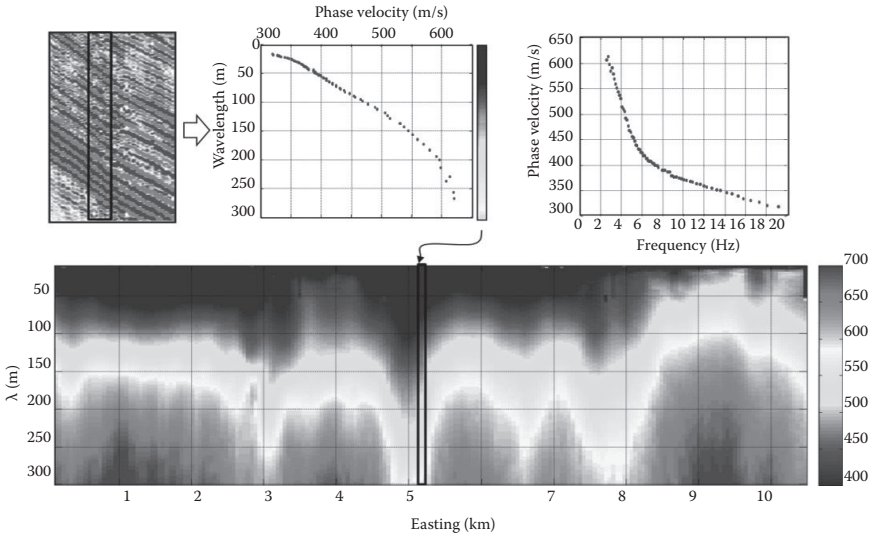


Figure 7.37 Dispersion curve for a location along a receiver line, plotted as phase velocity versus frequency, as phase velocity versus wavelength (top). Pseudosection of the phase velocity of the fundamental mode for a 10-km-long receiver line. (From Strobbia, C. et al., *First Break* 28, 85–91, 2010.)

For typical 3D geometries, the results of the analysis along source and receiver lines are merged into a volume representing the surface wave properties within a survey. The dispersion volume represents the phase velocity of a mode as a function of the geographic coordinates of the analysis point (easting and northing) and of the wavelength. The result of the first stage of analysis can be used to map the noise properties within the survey area. Due to the rapid changes from weathering, compaction, and lithology in the near-surface, large variations of the surface wave properties are often observed, even within a single receiver line. In Figure 7.38, a slice of the dispersion volume, at the wavelength 130 m, is depicted over a false-color satellite image for an area of about 300 km^2, from a 3D survey in Egypt (Strobbia et al. 2011).

The receiver lines are in the east–west direction and are spaced 210 m. The volume is obtained using about 30,000 dispersion curves. The fundamental mode is the highest energy event in a wide frequency range. Locally higher modes and P-guided waves are present.

The high lateral resolution of the surface waves allows mapping lithological boundaries and geological structures printing through the near-surface. For example, the high-velocity zones in the south correspond to wadi beds. The low-velocity zones in the southeast to northwest direction are fault zones, parallel to the main fault system (the Gulf of Suez, visible on the right). More details of the geological interpretation are provided by

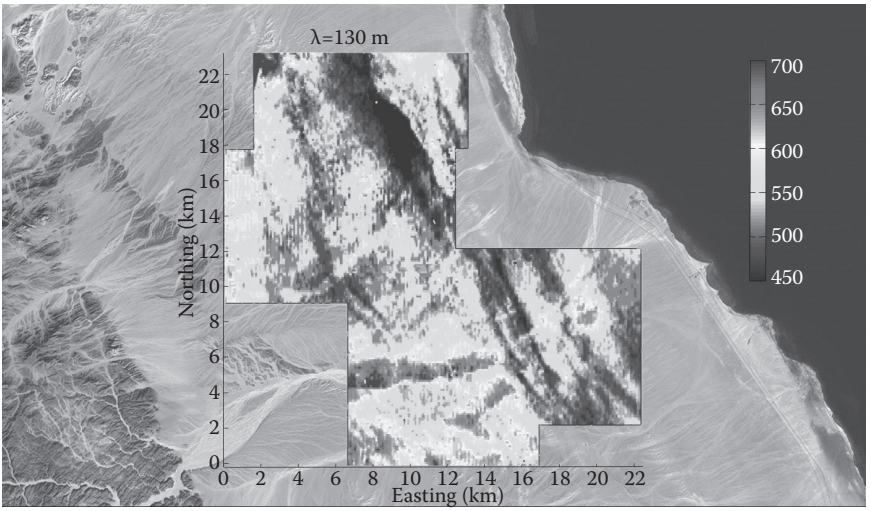


Figure 7.38 Wavelength slice of a dispersion volume for an area of about 300 km^2 showing the phase velocity of the Rayleigh waves for a wavelength of 130 m. (From Strobbia, C. et al., *Near Surf Geophys* 9, 503–514, 2011.)

(Laake et al. 2010). Mapping the surface wave properties of large areas allows the identification of large-scale geological features of the near-surface. The surface wave propagation properties can be merged with other data to build an integrated near-surface geological model. The identification of structural elements can be precious geological information, not only in frontier exploration projects.

The surface wave velocity has to be inverted to infer a near-surface model for data-processing applications. For each location, the adaptive surface wave analysis provides a unique set of local surface wave properties, which correspond to the local average properties. To maximize the lateral resolution in a case where the data quality is high, the inversion can be run with a local 1D approach. In some cases, 2D or 3D schemes are used. Essentially, they add a lateral regularization term because most 2D and 3D inversion approaches are still based on 1D forward modeling.

The quality and the lateral smoothness of dispersion volumes do not generally require further lateral smoothing or constraints. A laterally smooth inversion could be obtained directly from the dispersion volume. When boundaries and sharp lateral variations are present, a high-resolution image of the surface wave properties must be used, and sharpening can be suitable.

When a grid-based inversion is run, the vertical dimension (depth) is discretized with variable thickness layers. The maximum depth and the

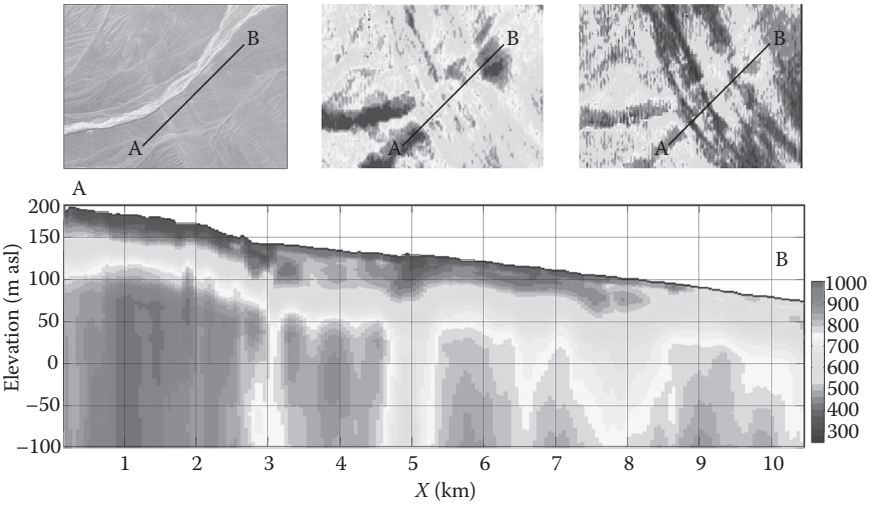


Figure 7.39 Shear wave velocity model obtained from the inversion of the pseudosection of Figure 7.37. The position of the section is shown on a satellite image and on two wavelength slices in the top part of the figure. (From Strobbia, C. et al., *Near Surf Geophys* 9, 503–514, 2011.)

cell size in the shallow section are the main model parameters. The near-surface model obtained via surface wave inversion provides geometric information about the near-surface layers, the geology, the velocity, and the attenuation distribution down to the investigation depth. With low-frequency sources and receivers, the investigated depth can reach hundreds of meters. In Figure 7.39, the results of a modal inversion are plotted for a receiver line, the pseudosection of which has been plotted in Figure 7.37. The model is a 2D velocity grid, with space variant cell size. The height of the cells increases with depth.

In geotechnical applications, the shear wave velocity is often the parameter of interest. On the contrary, in seismic reflection processing for velocity modeling and statics computation, a P-wave velocity model is required. The conversion from V_S to V_P can be done with lithological and hydrogeophysical information extracted from the near-surface integrated model. This can be relatively straightforward in arid areas with sediments where the Poisson ratio is known to have a very narrow range of values, and it becomes more critical in a context where the hydrogeophysical setting has important variations. Calibration with refraction data and up-hole or down-hole data allows the validation of the V_S to V_P conversion. Alternative approaches include the joint inversion of refraction travel times (Section 8.3.2.3).